



Masterarbeit

ScholarCopilot with Fine-Grained Citation Retrieval

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Niklas Munkes (Matrikelnummer 4269436), August 24, 2025

Abstract

TODO

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1 Introduction

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1.1 Retrieval-Augmented Generation

Retrieval-Augmented Generation (RAG, [LPP⁺21]) models are designed to generate text that is grounded in real-world factual knowledge. They have been widely adopted for knowledge-intensive NLP tasks [SQK⁺25] such as long-form question answering [XWL⁺24], scientific question answering [LOS⁺23] and question synthesis [XLS⁺24], generation with sentence-level citations [ZBL⁺24, XWL⁺24] and academic text generation [AHS⁺24, WMN⁺25]. RAG models usually consist of a generator, a pre-trained sequence-to-sequence (seq2seq) model (e.g. a Transformer-based [VSP⁺17] decoder, see section 2), and a retriever with access to a retrieval corpus, which may be implemented as a dense vector index (e.g. a HNSW [MY16] index, see section 3.1.1) [LPP⁺21]. Thus they combine pre-trained parametric memory with non-parametric (retrieval-based) memory [LPP⁺21].

The RAG architecture initially proposed by [LPP⁺21] consists of a query encoder q , a document index d and a generator p_θ (see Figure 1.1) TODO

By using an external retrieval corpus, the knowledge RAG models use to inform their generation can directly be inspected and interpreted [LPP⁺21]. Their TODO

citations build trust [DOW⁺24]

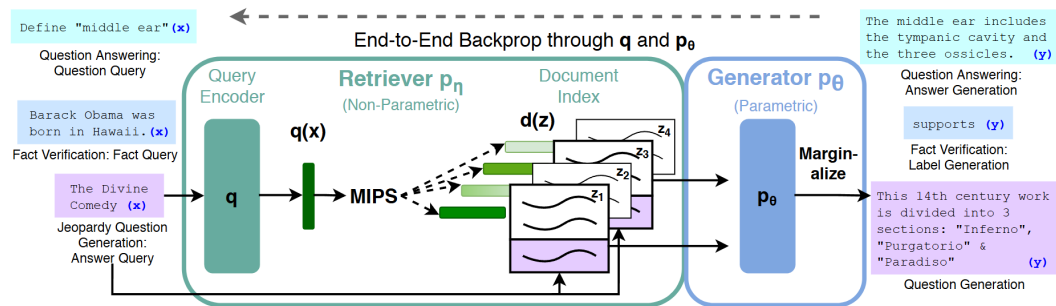


Figure 1.1: Overview of the RAG architecture [LPP⁺21]. For finding the top- k documents z_i , the authors used Maximum Inner Product Search (MIPS).

2 Related Work

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Previous work on text generation with citations has recently transitioned from treating retrieval and generation as separate processing stages (e.g. [SHC⁺24]) to alternating between the two (e.g. [XWL⁺24], [AHS⁺24]). *Attribute First, then Generate* [SHC⁺24] initially retrieves a set of relevant sub-sentence spans which are then grouped into coherent clusters which in turn are used to inform an iterative sentence-by-sentence generation step [SHC⁺24]. *ReClaim* [XWL⁺24], more specifically the *ReClaim with Interleaving Generation* method, is used to generate an output sequence $O = \{r_1, c_1, \dots, r_n, c_n\}$ of alternating references and claims where claim c_i is the portion of the models response that was generated based on reference r_i . Claim and reference generation is performed by separate LLMs specifically trained for their corresponding task [XWL⁺24]. In contrast, *OpenScholar* [AHS⁺24] first retrieves and reranks a set of papers P relevant to the given query. These papers inform a subsequent generation step that produces a generated response r_i [AHS⁺24]. This response is refined through iterative *self-feedback* where during each iteration of the feedback loop $F = f_1, \dots, f_T$, the generator model generates sentences f_j that describe potential improvements to the response and may also prompt the retrieval pipeline to retrieve additional papers, should r_i require more information to generate an improved response r_{i+1} [AHS⁺24]. Depending on the size of P and the choice of T , OpenScholar still performs retrieval and generation largely separately.

ScholarCopilot [WMN⁺25] is a RAG system developed for generating academic-style text with accurate citations. It is intended as a tool that assists the user with academic paper writing by providing iterative text generation with automated citation retrieval [WMN⁺25]. ScholarCopilot also enables user interaction during generation [WMN⁺25] by allowing the user to rewrite the generated text or forcefully trigger citation retrieval between the consecutive generation steps but this feature is beyond the scope of this thesis and is thus not utilized for the experiments. ...

TODO

3 Methods

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3.1 ScholarCopilot

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3.1.1 HNSW

TODO

3.1.2 ScholarCopilot Training

TODO

3.2 Passage Reranking

TODO

3.2.1 Batched Self-Consistency

TODO

3.2.2 Min-Max Normalization

TODO

3.3 Deepseek R1

TODO

4 Experiments

TODO

4.1 Retrieval Evaluation

TODO

4.2 Generation Evaluation

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4.3 Additional Experiments

TODO

5 Limitations and Future Work

TODO

6 Conclusion

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